PROGRAMMING FUNDAMENTALS · ASSIGNMENT 02

Before You Start

From the sample-files ZIP to your first commit, step by step, with Visual Studio and GitHub

Worked example account

`github.com/L1F24BSCS0081`

Deadline: Thursday, 29 October 2026 · 11:55 PM

The whole journey at a glance

1

Extract the ZIP

Get the dummy input files and starter code out of the ZIP.

2

Create a project

One Visual Studio Empty Project per question, with the exact .cpp name.

3

Copy the .txt files

Into the project folder, next to the .vcxproj file.

4

Write and run

Ctrl + F5, then compare with the sample output line by line.

5

Test other data

Change values, delete lines, rename the file, empty it.

6

Upload to GitHub

One .cpp per commit, with the commit messages from Task A3.

7

Share and submit

Invite the instructor, then upload ONE PDF on the portal.

Golden rule

The .txt file lives in the same folder as the program.

Your folder plan: one project per question

```
Assignment02\
├─ Assignment02_Sample_Files\
├─ Q1_stock_debug\
├─ Q2_parking_report\
├─ Q3_scholarship\
├─ Q4_heart_rate\
└─ Q5_expense_tracker\
```

Put Assignment02 anywhere you like. The first folder inside comes from the ZIP; Visual Studio creates the other five.

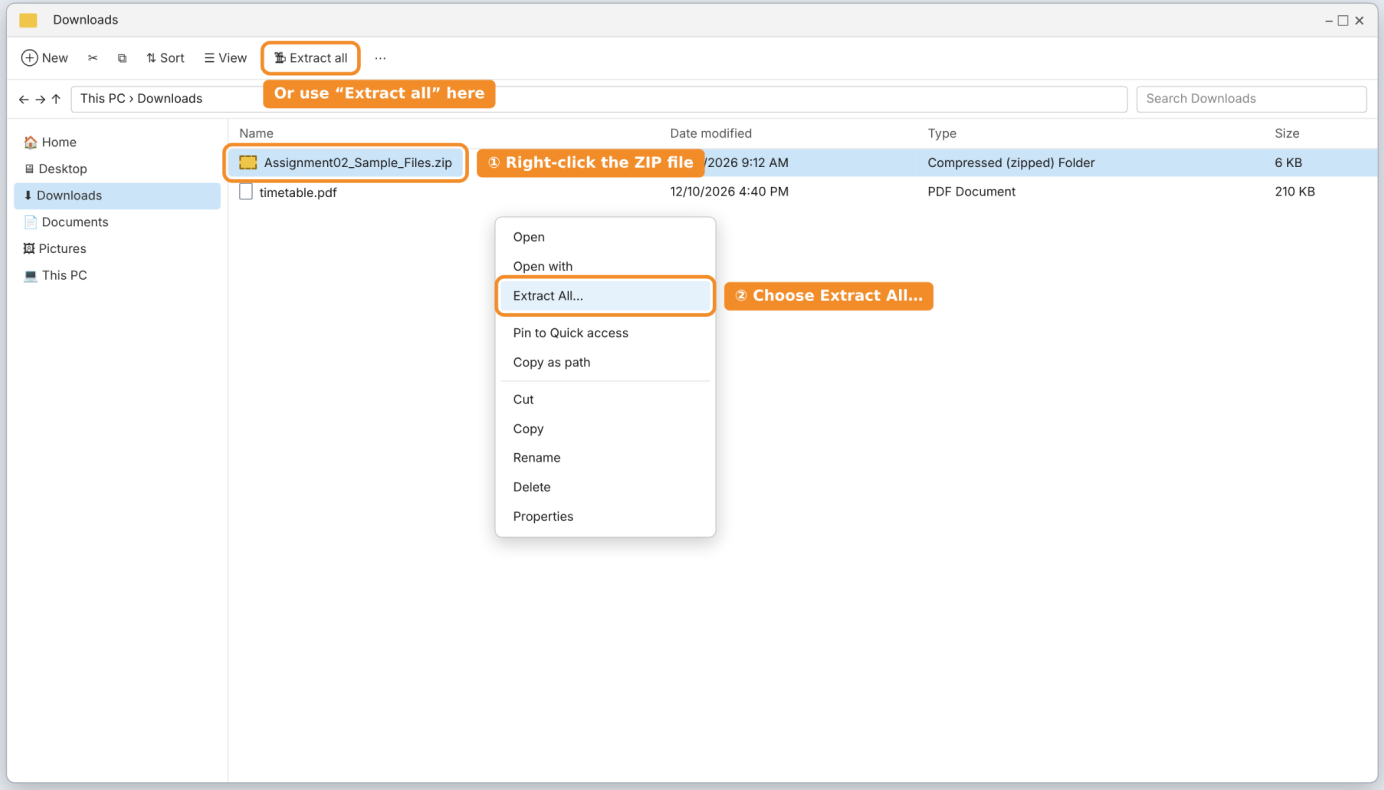
Q	Project name	Copy into its folder	Your program creates
Q1	Q1_stock_debug	Q1_stock_debug.cpp, stock.txt, reorder.txt	adds lines to reorder.txt
Q2	Q2_parking_report	parking.txt	receipts.txt
Q3	Q3_scholarship	class1.txt and the “my files” folder	scholarship.txt
Q4	Q4_heart_rate	Q4_heart_rate.cpp, heart_rate.txt	clean_heart_rate.txt
Q5	Q5_expense_tracker	nothing — no input file	expenses.txt

STEP 1

Download and extract the ZIP

1. Download **Assignment02_Sample_Files.zip** from the course portal.
2. Open your **Downloads** folder and right-click the ZIP file.
3. Choose **Extract All...** (or click Extract all in the toolbar).

Never work inside the ZIP. Files opened straight from a ZIP are temporary copies, and your program will not find them.



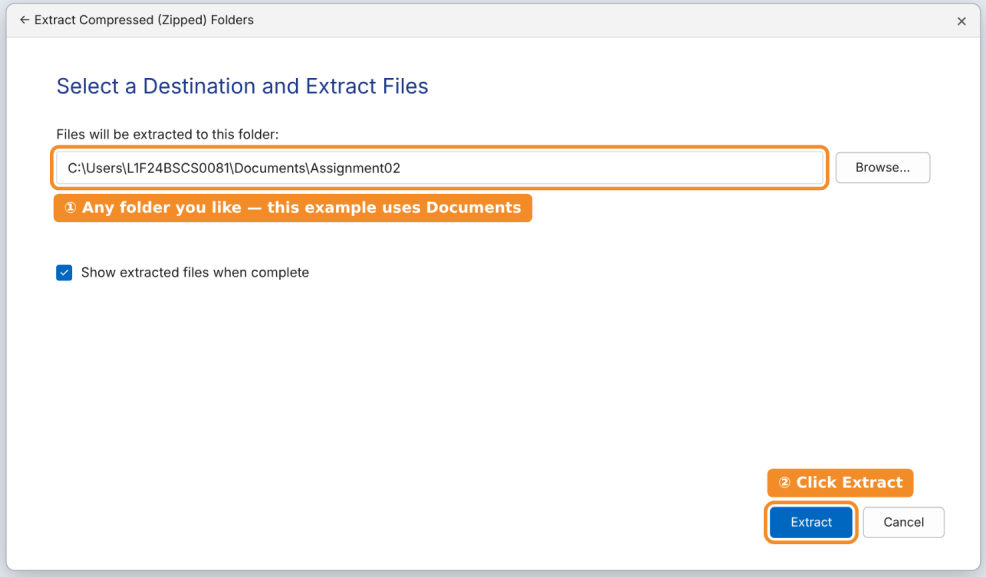
Illustrative mock-up — the real screens may look slightly different.

STEP 1

Extract into one Assignment02 folder

1. Click **Browse…**, pick **any place you like** (Desktop, Documents, D: drive) and make a folder named **Assignment02**.
2. Select that folder. The example on the right happens to use **Documents**.
3. Click **Extract**. A folder named Assignment02_Sample_Files appears inside.

Why here? Your five Visual Studio projects will go in this same folder, so copying files between them is easy.



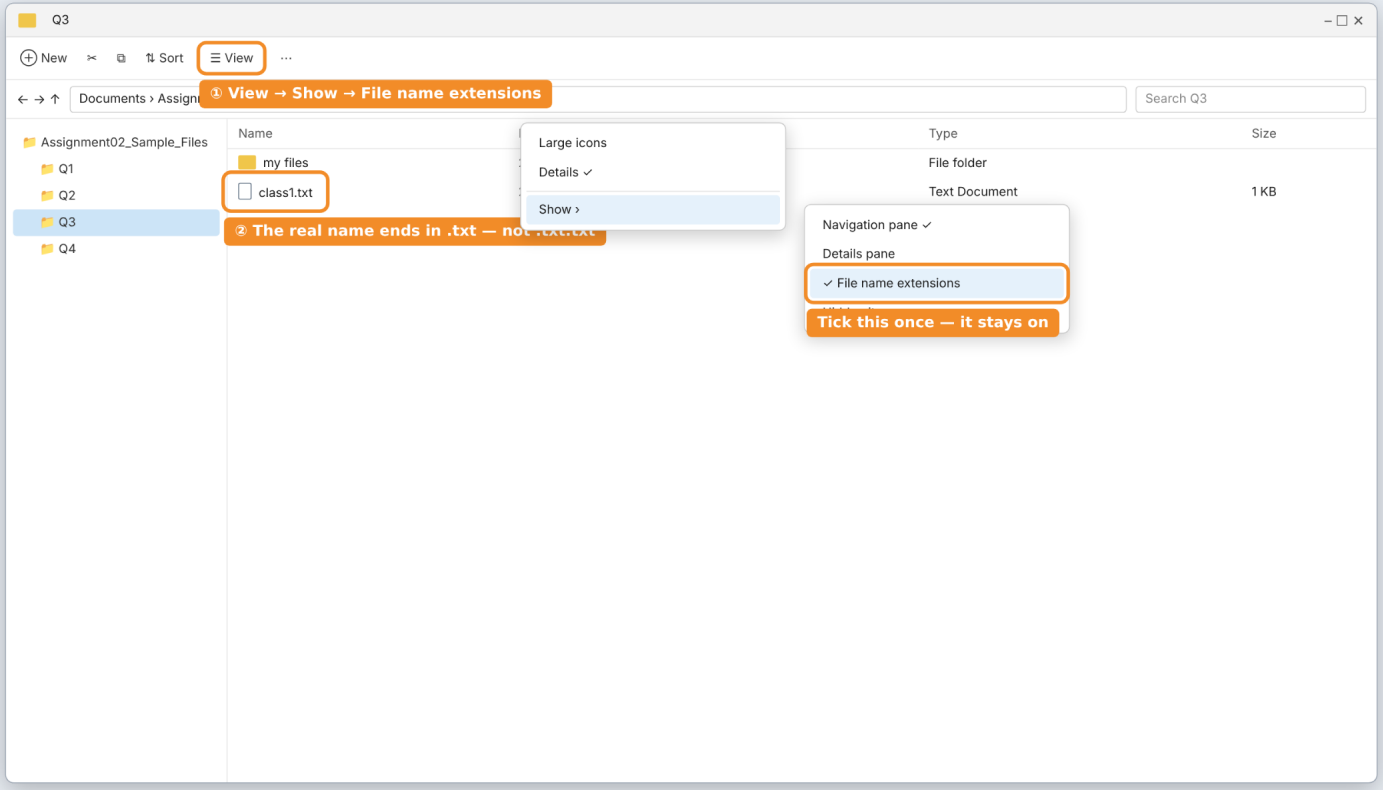
Illustrative mock-up — the real screens may look slightly different.

STEP 1

Show the real file names

1. In File Explorer, open **View → Show**.
2. Tick **File name extensions**. You only need to do this once.
3. Every file now shows its ending, for example **class1.txt**.

See parking.txt.txt? Rename it to parking.txt, otherwise your program prints “parking.txt not found!”.



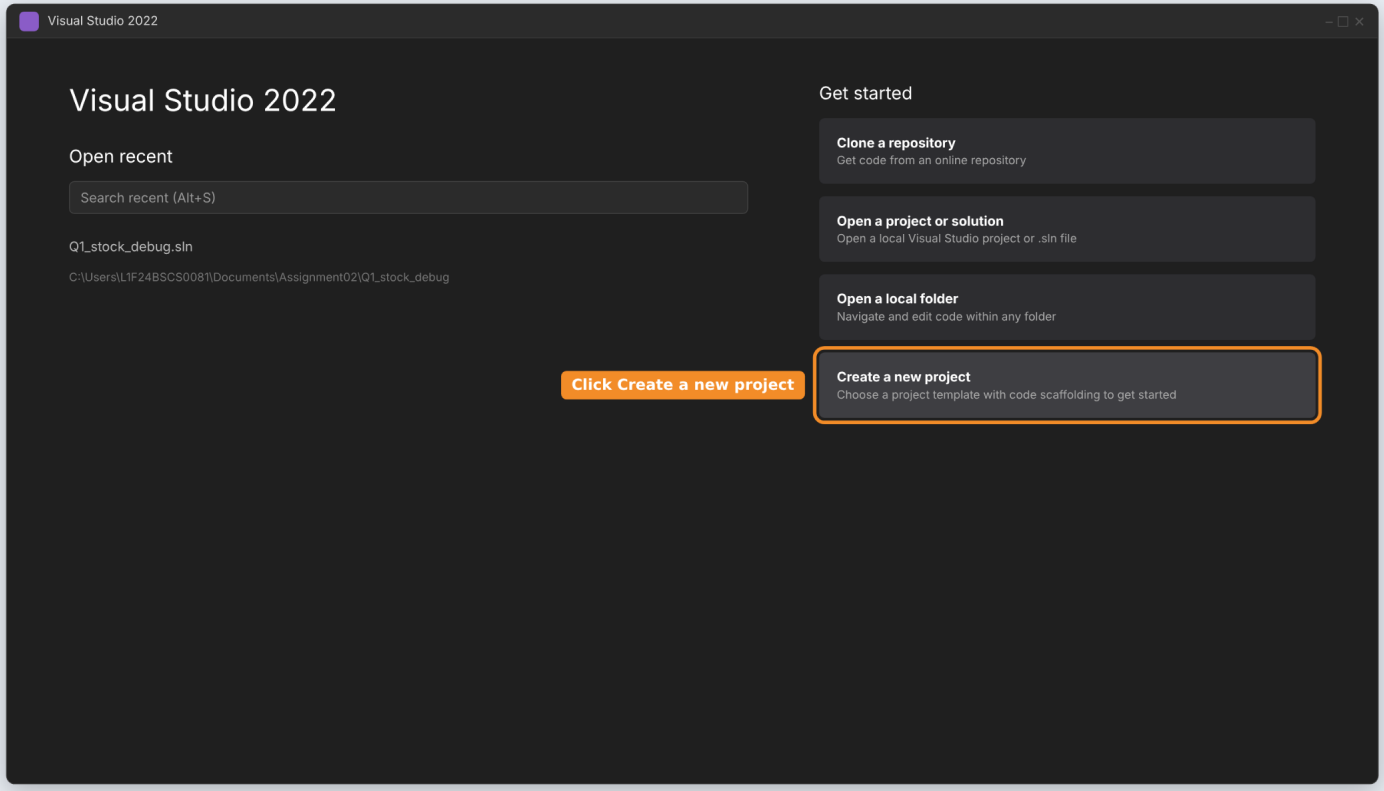
Illustrative mock-up — the real screens may look slightly different.

STEP 2

Start a new Visual Studio project

1. Open **Visual Studio 2022**.
2. On the start window, click **Create a new project**.
3. Repeat Steps 2 to 4 for every question: **five projects** in total.

Already in Visual Studio? Use File → New → Project instead.



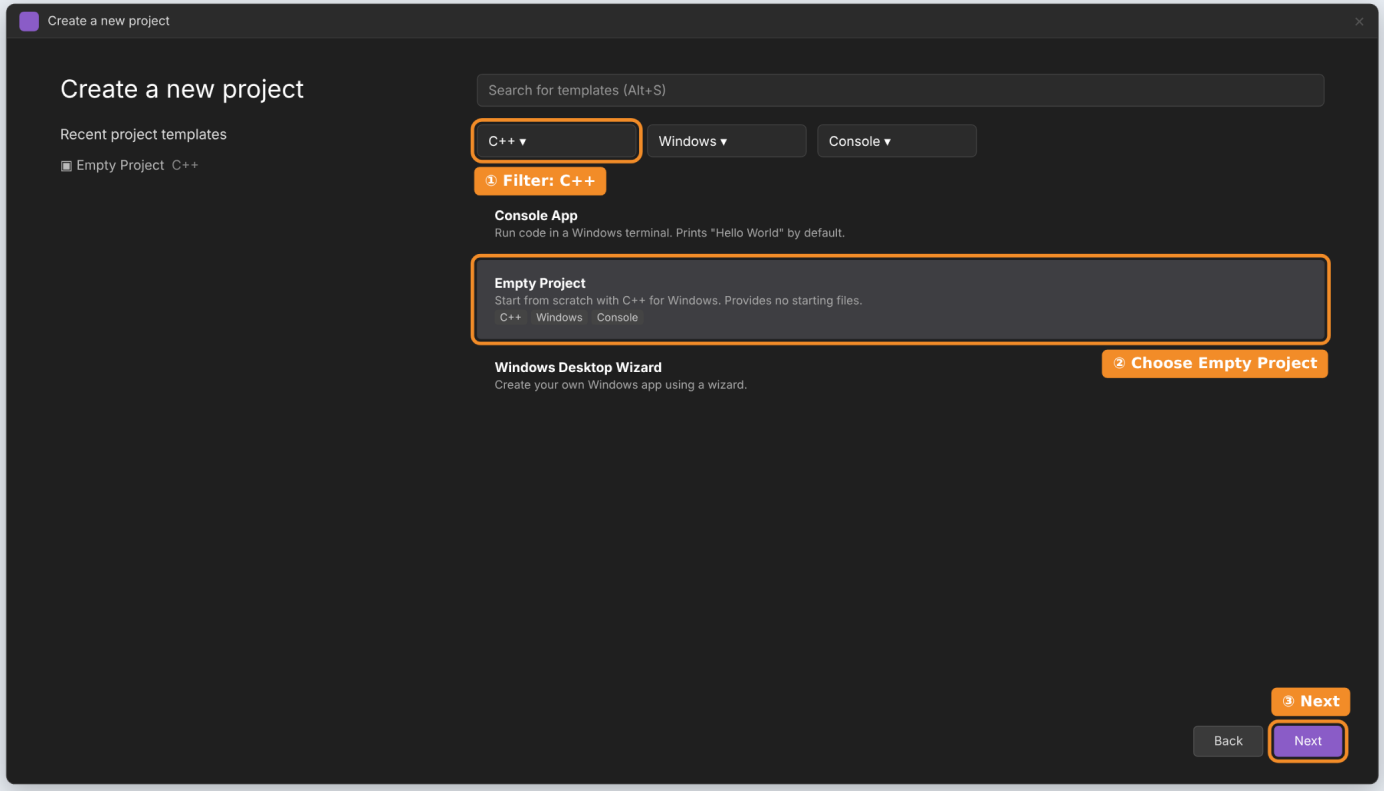
Illustrative mock-up — the real screens may look slightly different.

STEP 2

Choose the Empty Project template

1. Set the language filter to **C++**.
2. Select **Empty Project**, not Console App.
3. Click **Next**.

Why not Console App? It creates a “Hello World” .cpp with the project’s own name. You would have to delete and rename it.



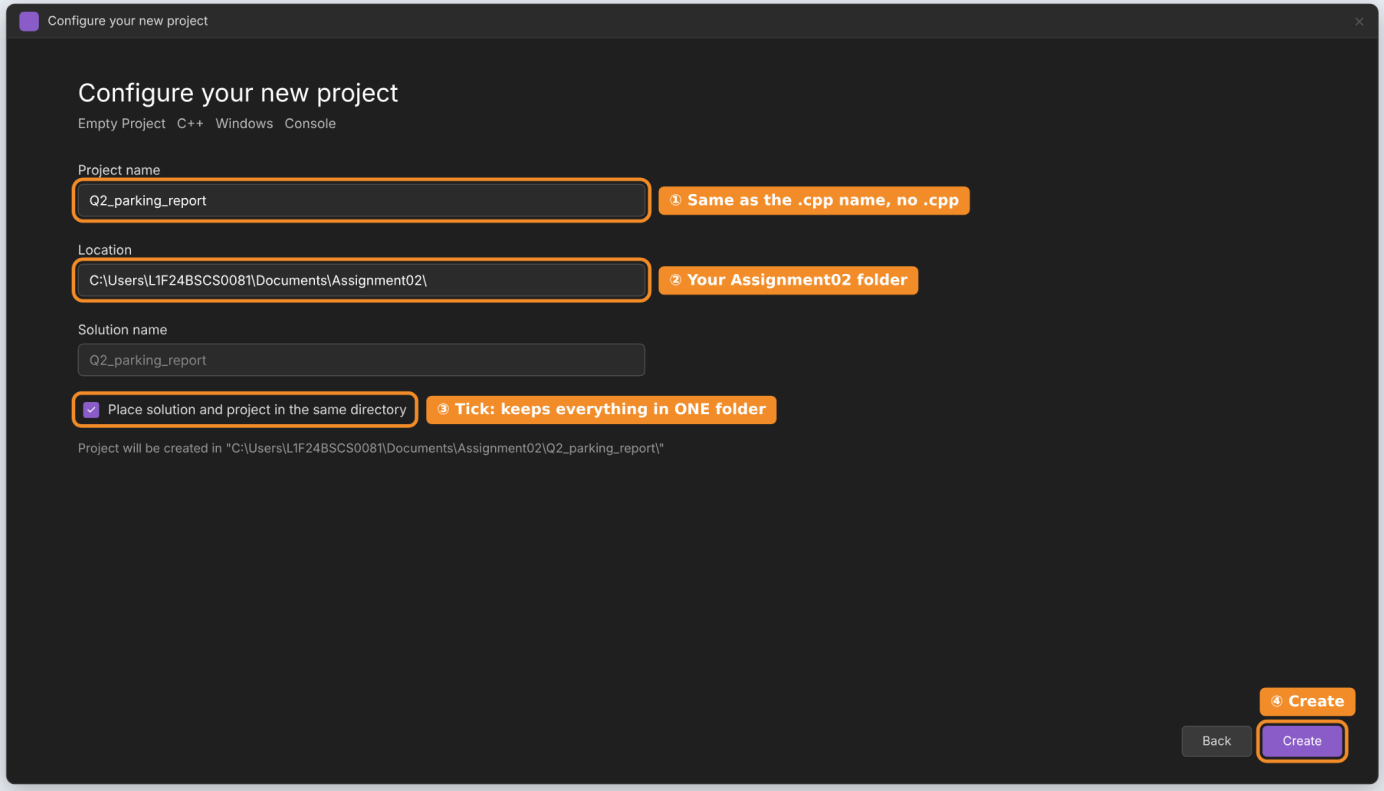
Illustrative mock-up — the real screens may look slightly different.

STEP 2

Name the project after the question

1. Project name: **Q2_parking_report**, the .cpp name without .cpp.
2. Location: the **Assignment02** folder you made in Step 1, wherever you put it.
3. Tick **Place solution and project in the same directory**.
4. Click **Create**.

The tick matters: it puts everything in ONE folder, so there is only one place for your .txt files.



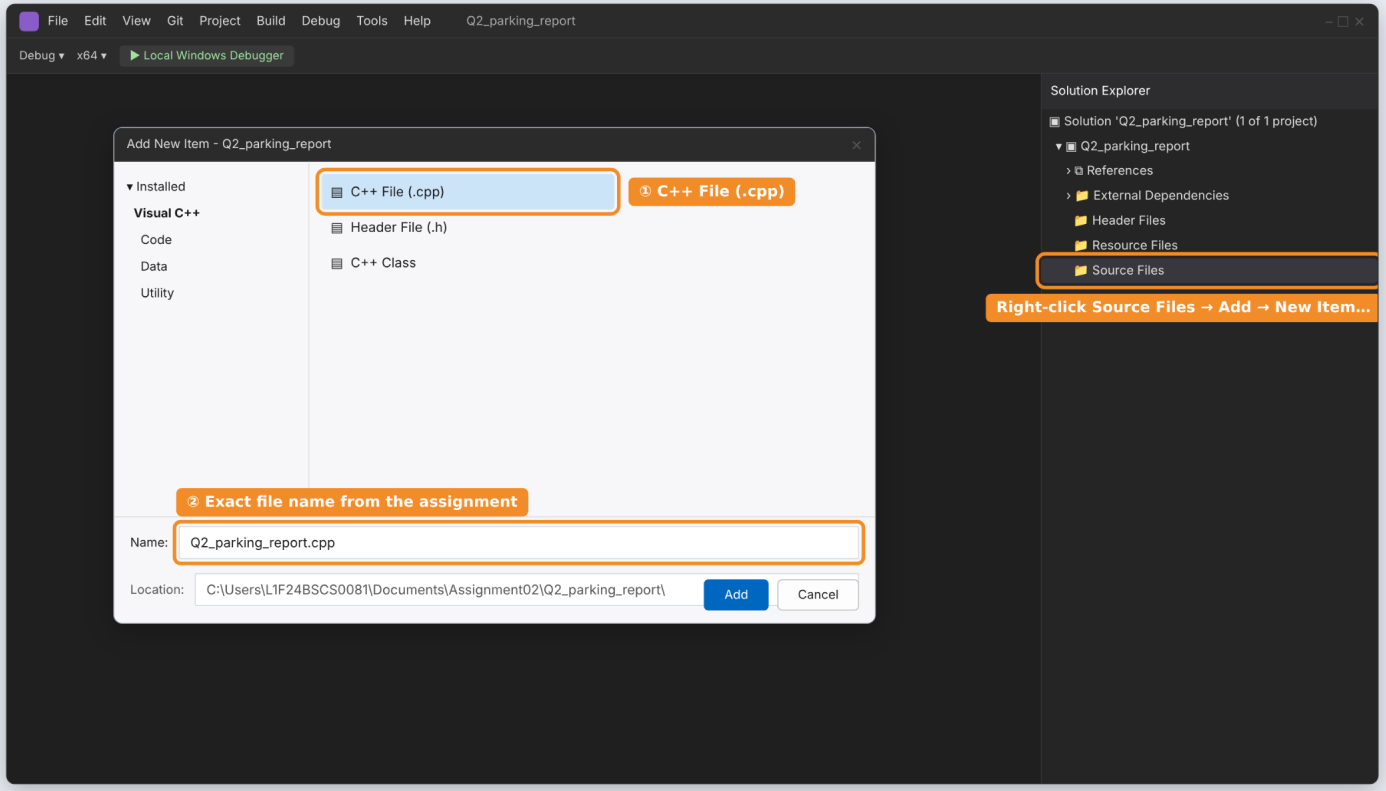
Illustrative mock-up — the real screens may look slightly different.

STEP 2

Add the .cpp file with the exact name

1. In Solution Explorer, right-click **Source Files → Add → New Item…**
2. Choose **C++ File (.cpp)**.
3. Name: **Q2_parking_report.cpp**, exactly as in the assignment.
4. Click **Add**.

Capitals count: q2_Parking_Report.cpp is a different name and is treated as not submitted.



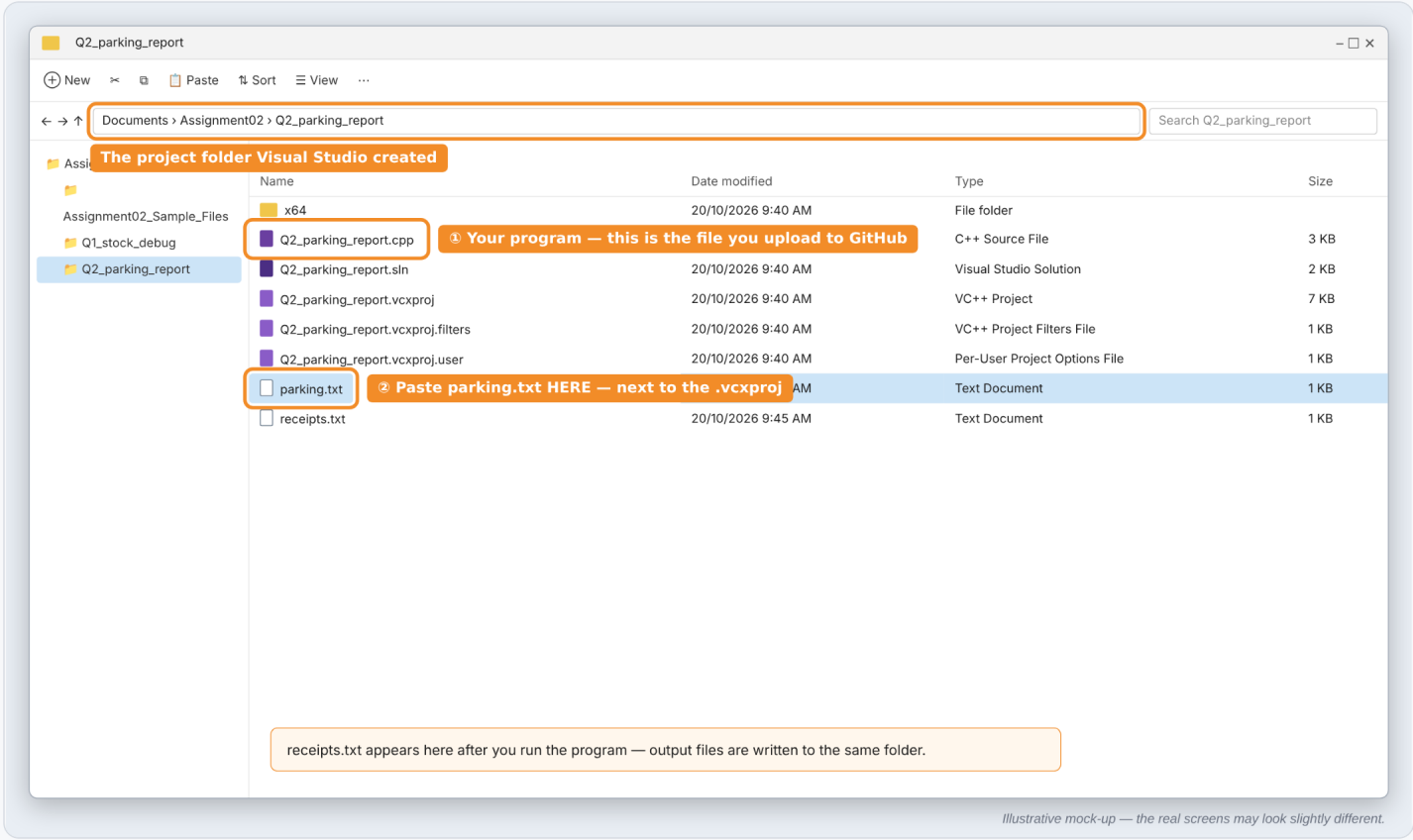
Illustrative mock-up — the real screens may look slightly different.

STEP 3

Put the .txt file in the project folder

1. In Solution Explorer, right-click the project → **Open Folder in File Explorer**.
2. Copy **parking.txt** from Assignment02_Sample_Files\Q2 and paste it here.
3. It must sit **next to the .vcxproj** file, not inside x64 or Debug.

Output files land here too. After a run, open receipts.txt from this same folder to check it.



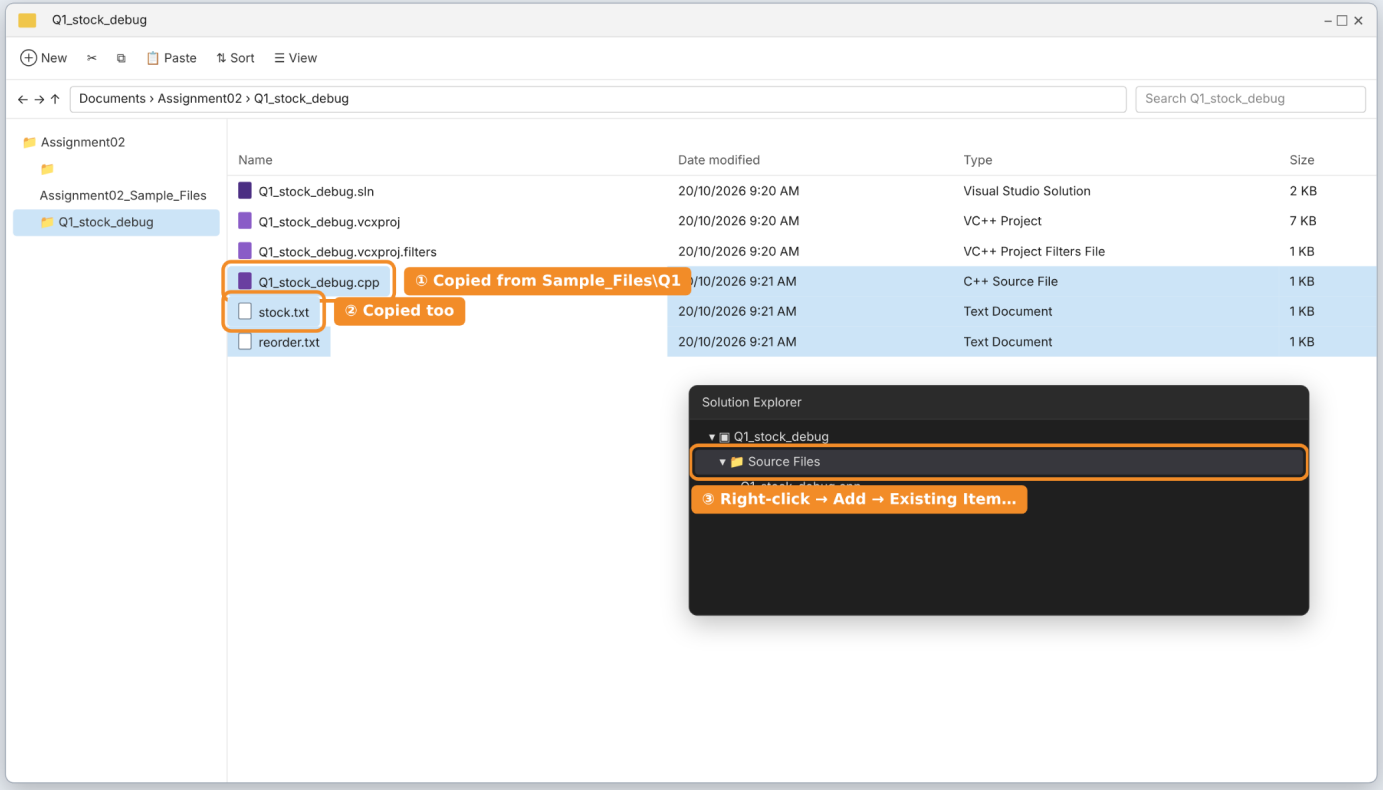
Illustrative mock-up — the real screens may look slightly different.

STEP 3

Q1 and Q4: start from the given code

1. Copy **Q1_stock_debug.cpp**, **stock.txt** and **reorder.txt** into the Q1_stock_debug folder.
2. Right-click **Source Files → Add → Existing Item…** and pick Q1_stock_debug.cpp.
3. Do the same for Q4 with **Q4_heart_rate.cpp** and **heart_rate.txt**.

One project, one .cpp. Do not also add a new empty .cpp to these projects.



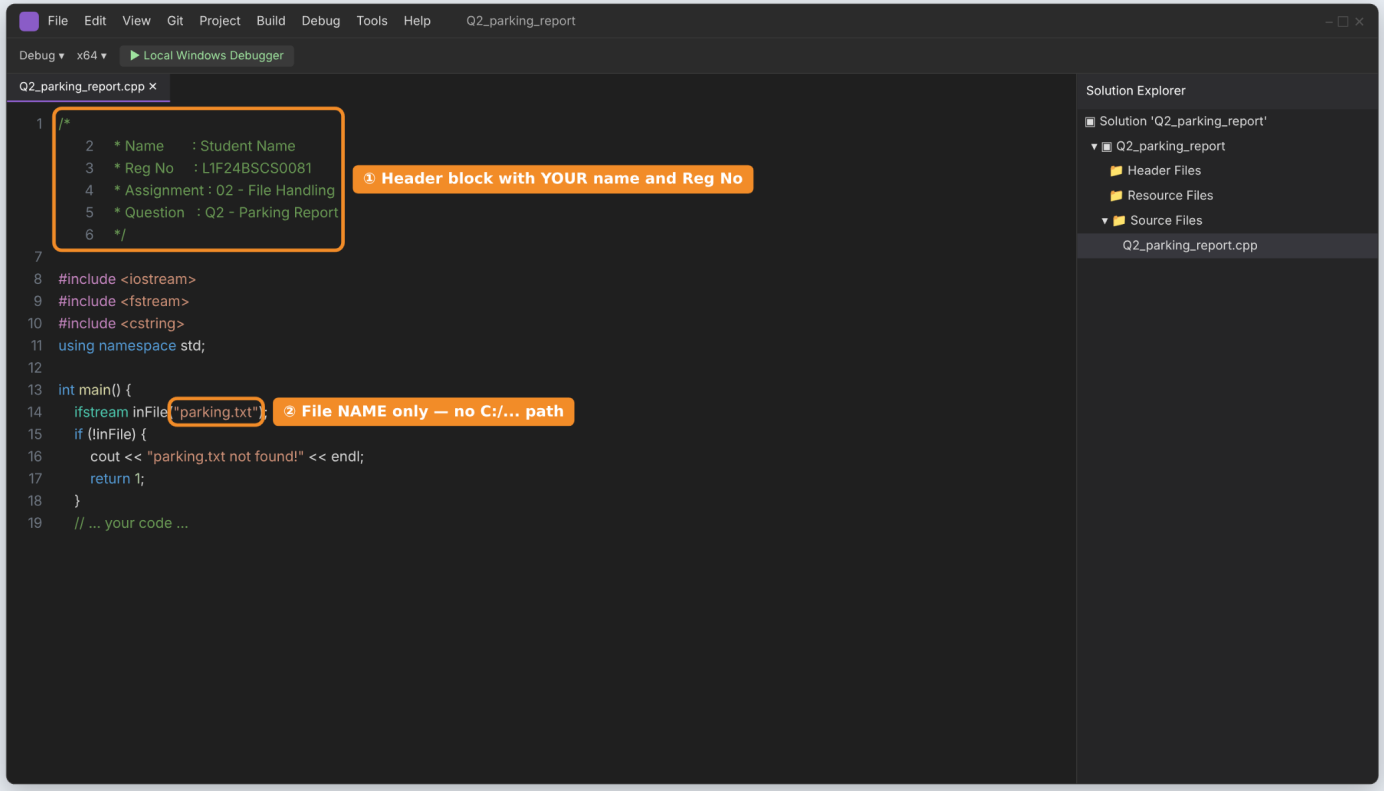
Illustrative mock-up — the real screens may look slightly different.

STEP 4

Write your program

1. Start with the **header comment block**: your name and Reg No.
2. Open files by **name only**: "parking.txt", never "C:/Users/...".
3. Use only **iostream**, **fstream** and **cstring**.

Build it in small steps: first read and print the file, then add the calculations, then write the output file.



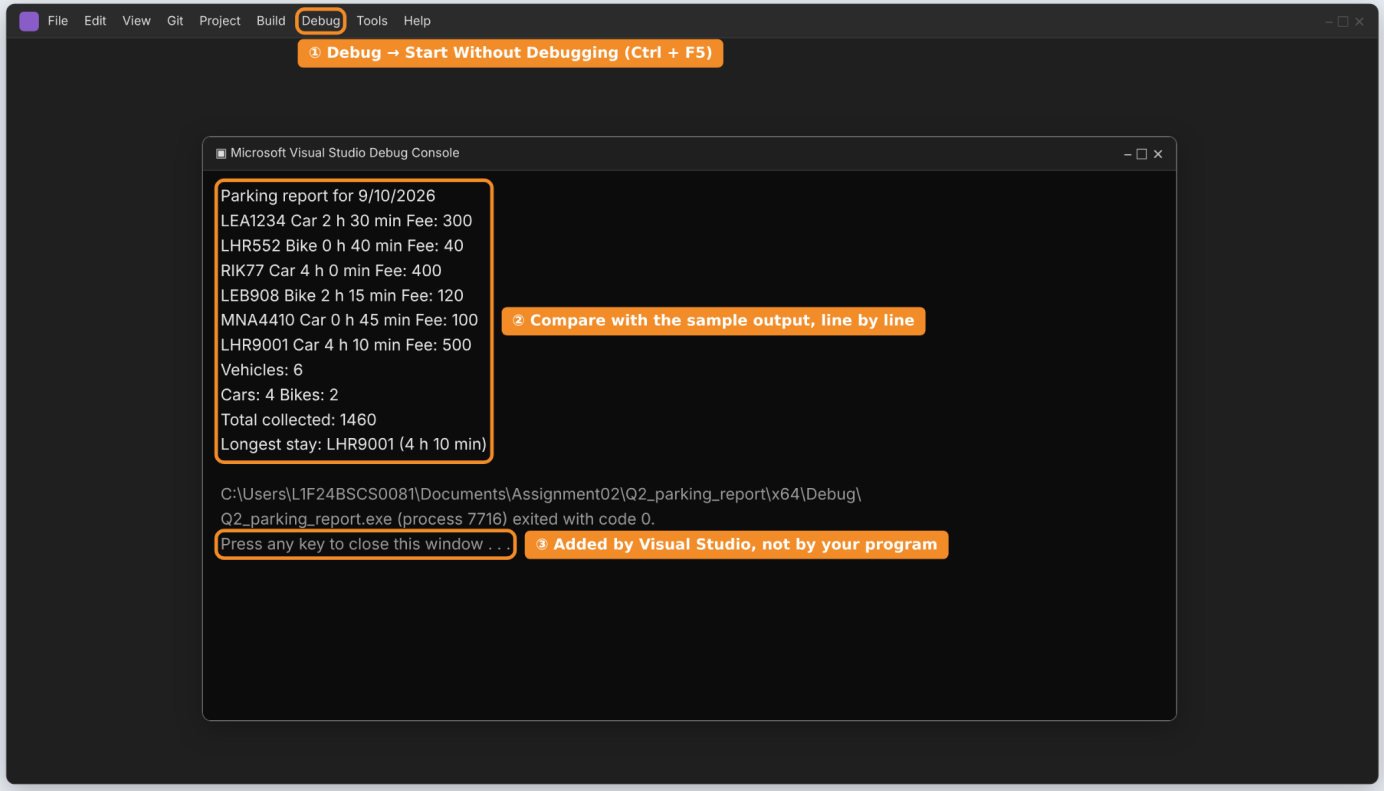
Illustrative mock-up — the real screens may look slightly different.

STEP 4

Run it and compare the output

1. Press **Ctrl + F5** (Debug → Start Without Debugging).
2. Compare **every line** with the sample output in the assignment.
3. "Press any key to close this window" comes from Visual Studio. Never add **system("pause")**.

Every character counts: spelling, capital letters, spaces between words and punctuation.



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STEP 4

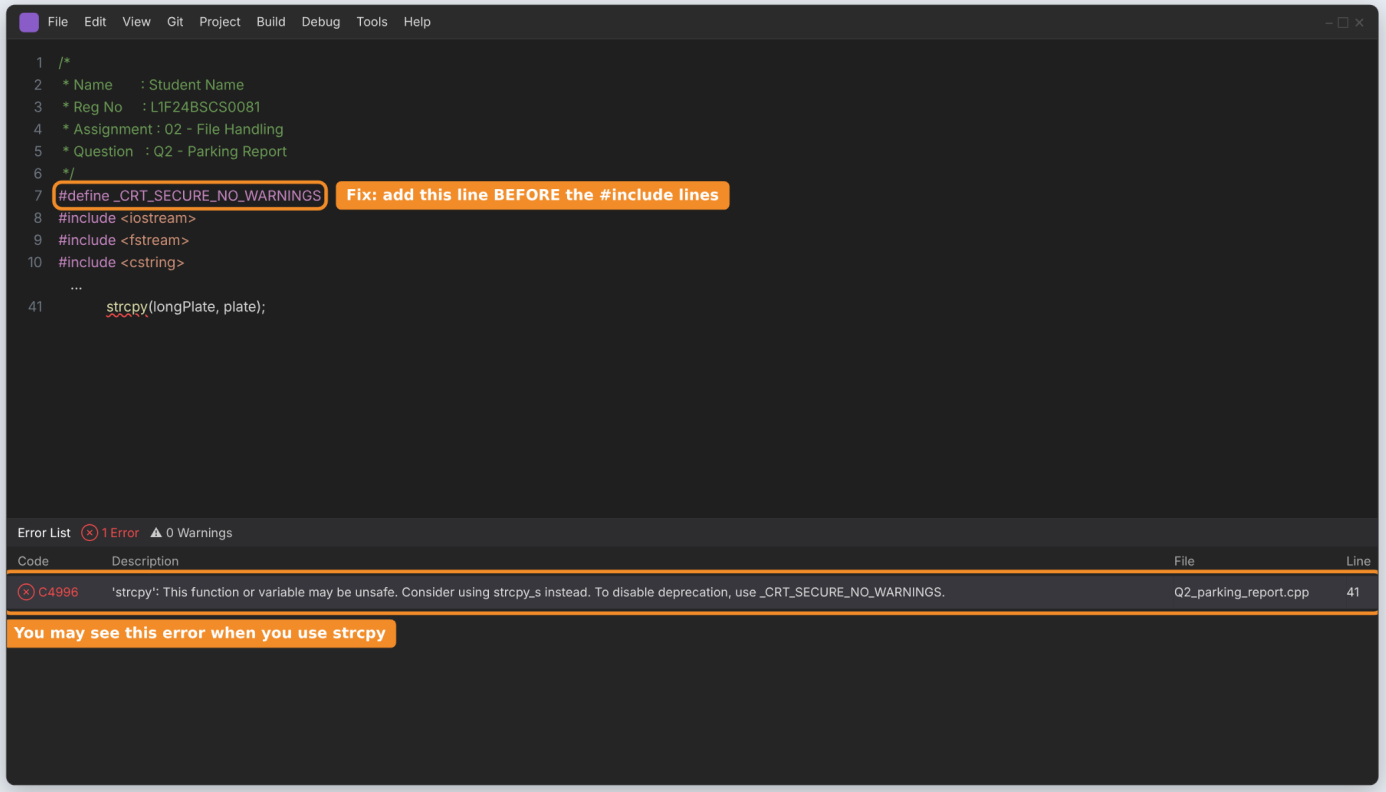
Error C4996 when you use strcpy

1. Visual Studio may refuse to build when Q2, Q3 or Q5 uses **strcpy**.
2. Add this line after the header comment, **before the #include lines**:

```
#define _CRT_SECURE_NO_WARNINGS
```

3. Build again and keep using **strcpy**, as taught in class.

Do not switch to strcpy_s. It is not standard C++ and may not compile elsewhere.



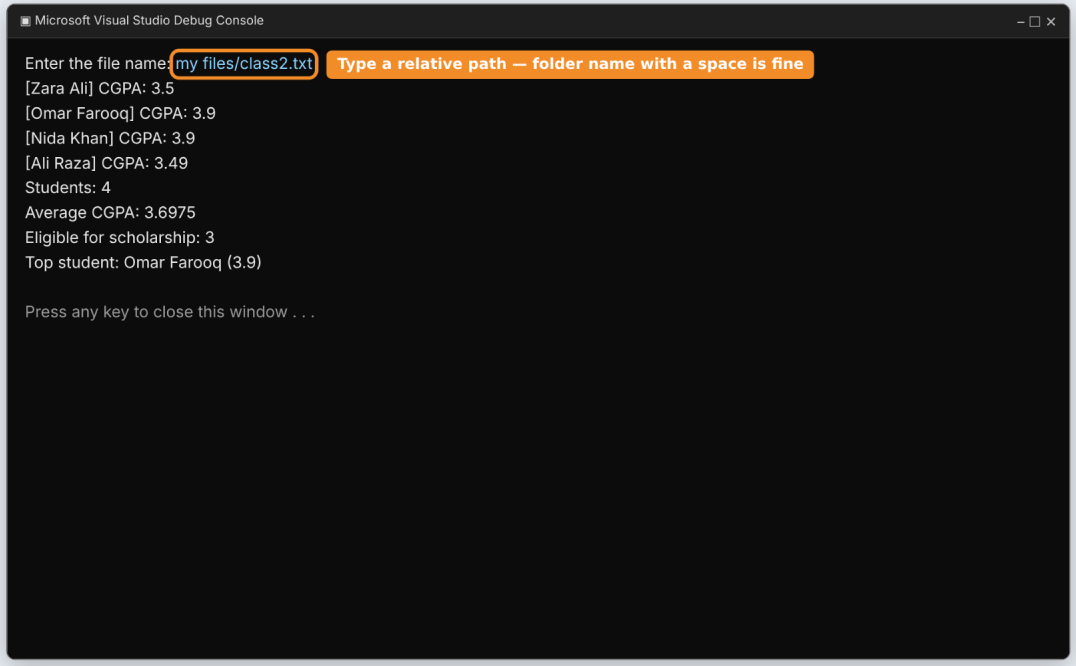
Illustrative mock-up — the real screens may look slightly different.

STEP 4

Q3: typing a file name or a path

1. Copy **class1.txt** and the whole **my files** folder into the Q3_scholarship folder.
2. Run and type **class1.txt**. Run again and type **my files/class2.txt**.
3. Run once more and type a wrong name, such as **class9.txt**.

Only Q3 asks for a path. The space in “my files” is why the name is read with cin.getline, not cin >>.



Illustrative mock-up — the real screens may look slightly different.

STEP 5

Test with your own dummy data

Your programs will also be checked with other files in the same format, so the sample file alone is not enough. Run each test below.

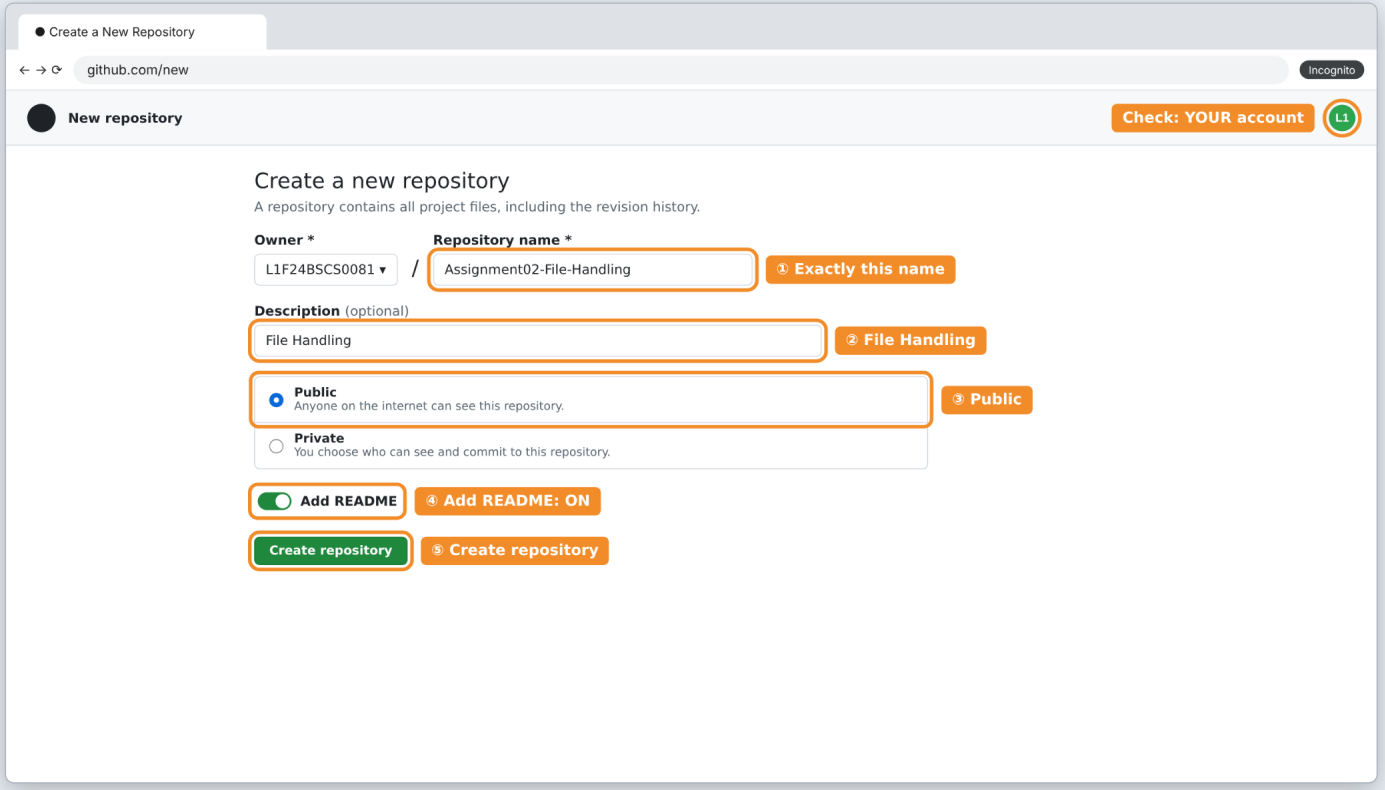
Sample file	Run with the .txt from the ZIP	Exactly the sample output
Other values	Edit the .txt in Notepad: change numbers, add or delete lines	Correct new results
Missing file	Rename the .txt (for example parking_old.txt) and run	The exact error message
Empty file	Delete everything in the .txt, save, and run	The empty-file message
Q5 restart	Add expenses, choose 5, run again, choose 2	The old expenses are still listed

After testing, put the sample files back as they were (Q1: reorder.txt back to its one original line).

STEP 6

Create the GitHub repository

1. Sign in to your **Assignment 1 account** (Incognito on lab PCs) and click **+ → New repository**.
2. Name: **Assignment02-File-Handling**
3. Description: **File Handling**. Visibility: **Public**. Add README: **ON**.
4. Click **Create repository**. This is commit 1.



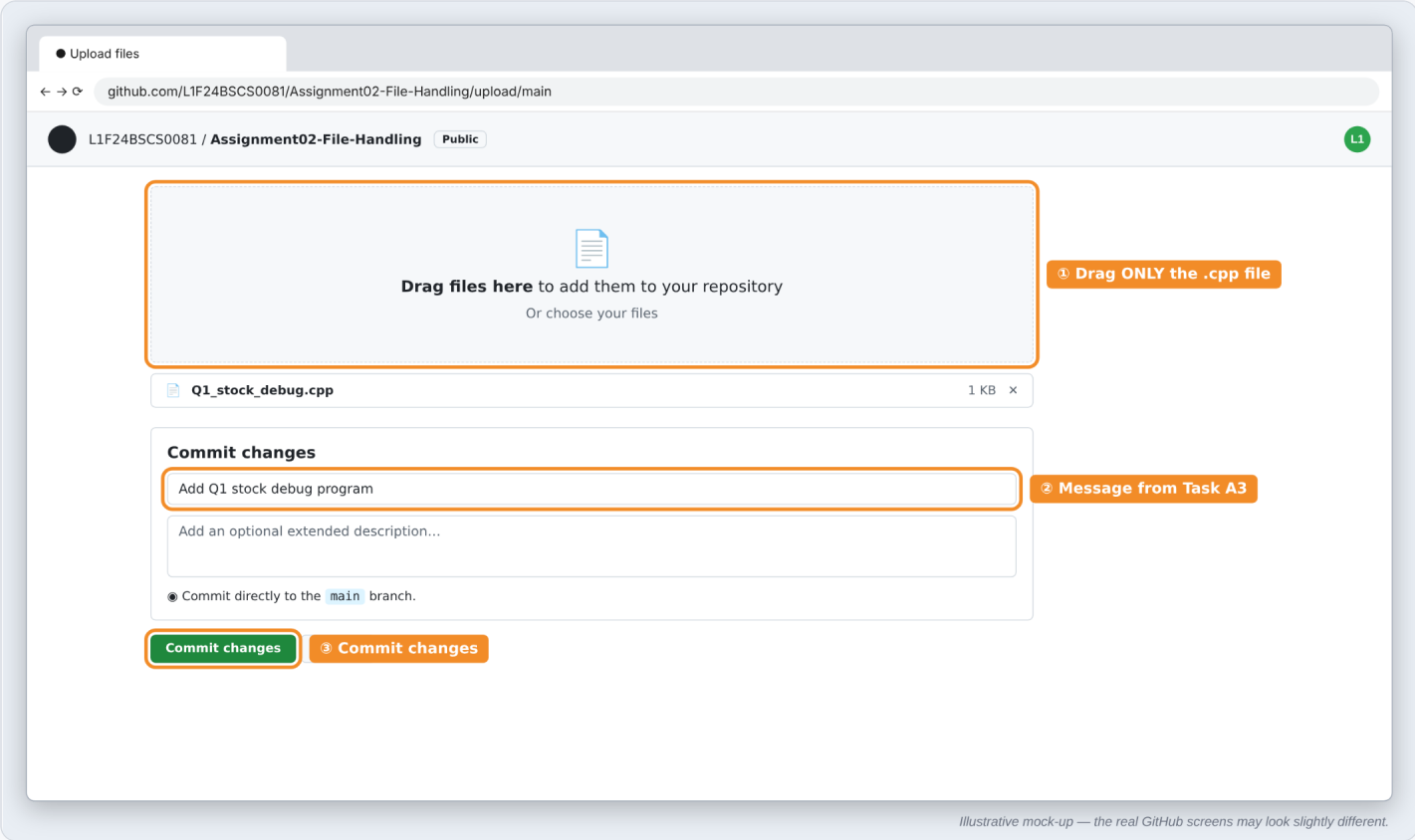
Illustrative mock-up — the real GitHub screens may look slightly different.

STEP 6

Upload each program as its own commit

1. On the repository page click **Add file → Upload files**.
2. Drag **only the .cpp** file from the project folder.
3. Commit message: **Add Q1 stock debug program**, then Commit changes.
4. Repeat for Q2 to Q5, **one file per upload**, with the messages from Task A3.

Never upload .sln, .vcxproj, the x64 folder or .exe files.



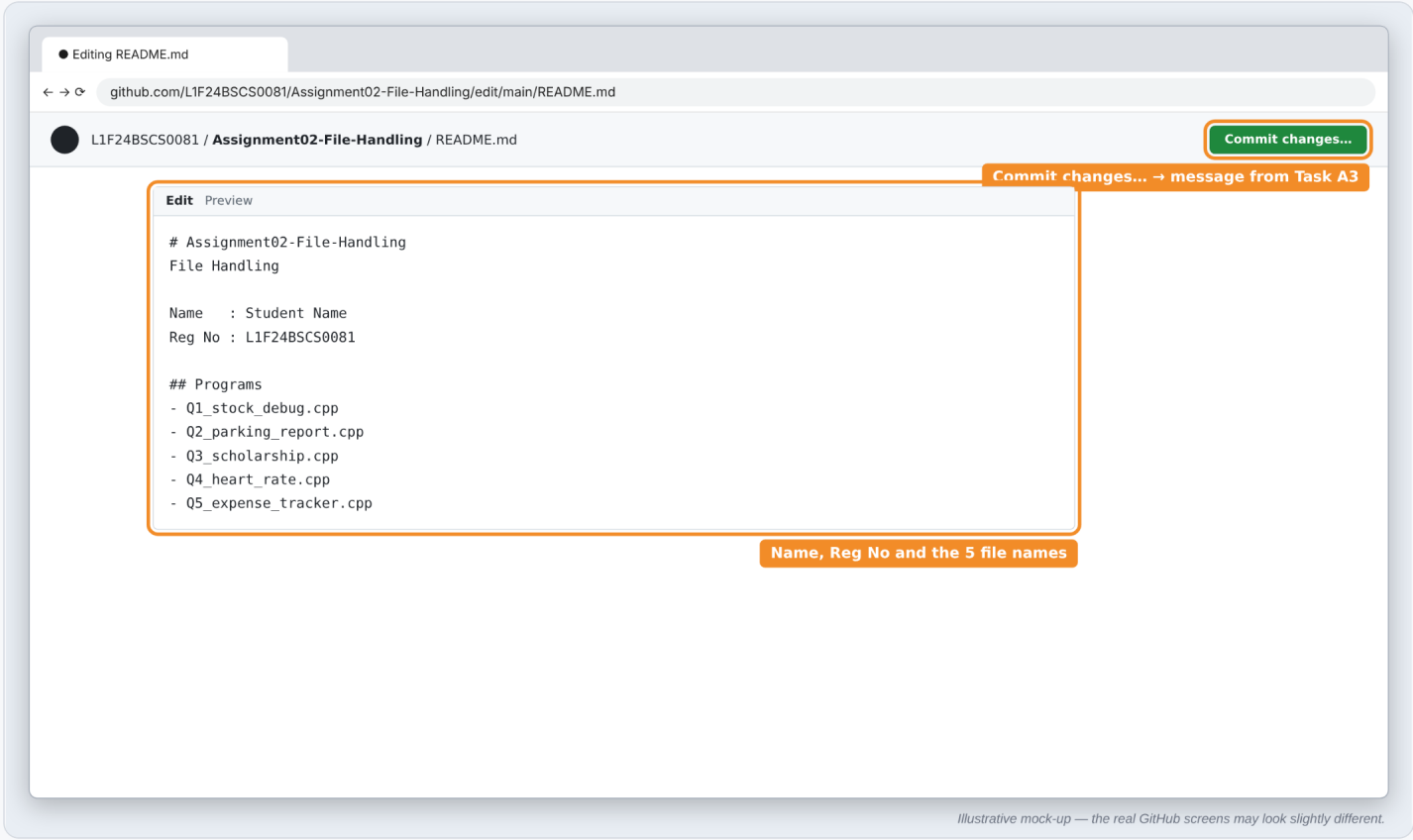
Illustrative mock-up — the real GitHub screens may look slightly different.

STEP 6

Update the README

1. Click **README.md**, then the pencil (**Edit**) icon.
2. Add your **full name**, your **Reg No** and the **five .cpp file names**.
3. Click **Commit changes…** with the message:

```
Update README with name, reg no
and question list
```



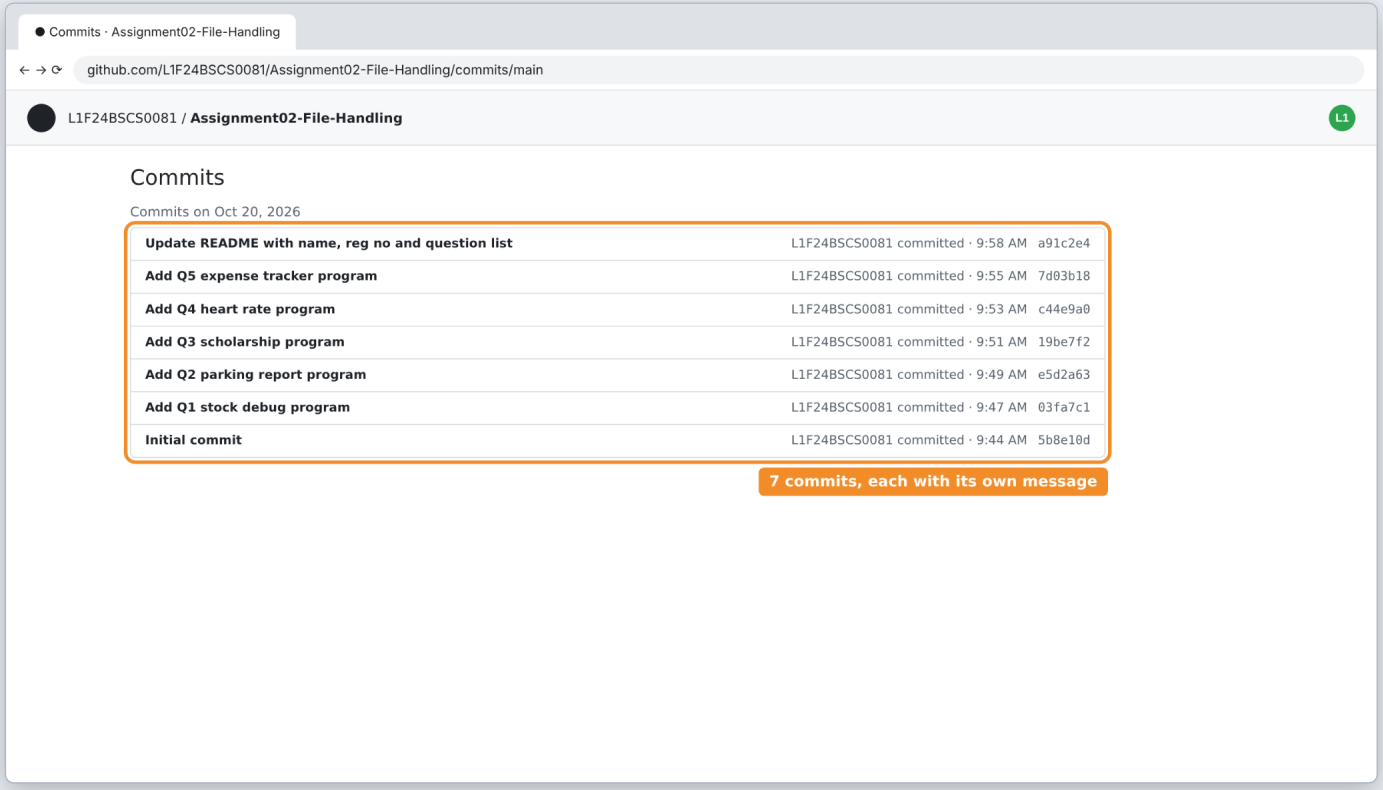
Illustrative mock-up — the real GitHub screens may look slightly different.

STEP 6

Check your commit history

1. On the repository page click the **Commits** count.
2. You should see **at least 7 commits**, all before **29 Oct, 11:55 PM**.
3. Take a full-screen screenshot for **page 2** of your PDF.

Fixed a program later? Upload it with the same name and a clear message, such as “Fix fee calculation in Q2”.

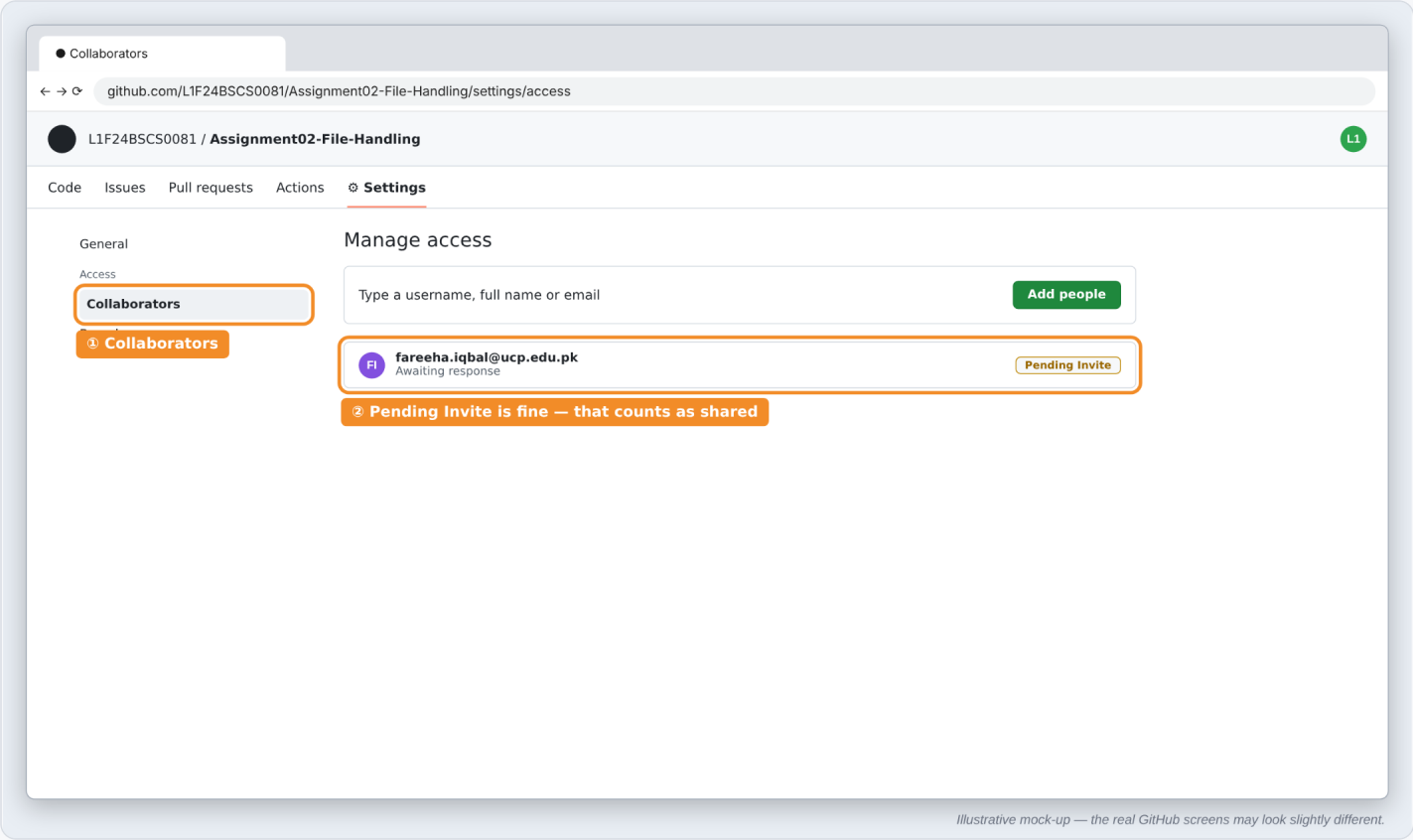


Illustrative mock-up — the real GitHub screens may look slightly different.

STEP 7

Invite the instructor

1. Open **Settings** → **Collaborators** and click **Add people**.
2. Type the email exactly: **fareeha.iqbal@ucp.edu.pk**
3. **Pending Invite** is fine. Screenshot this page for **page 3** of your PDF.



Illustrative mock-up — the real GitHub screens may look slightly different.

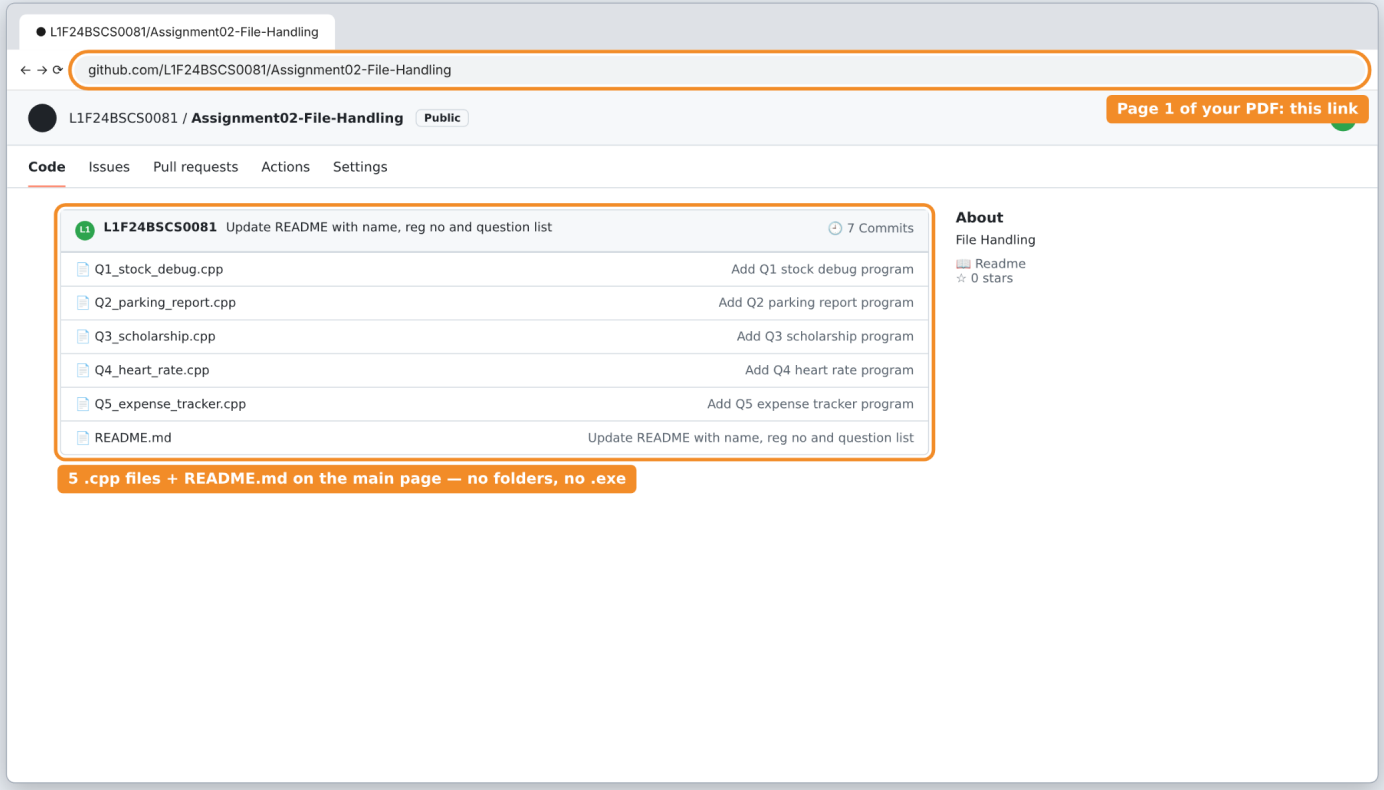
STEP 7

Final check, then submit the PDF

1. The repository page shows **5 .cpp files + README.md**, no folders.
2. Copy the link from the address bar for **page 1** of your PDF.
3. Make **ONE PDF** (link, commits, collaborators) named with your Reg No.

```
L1F24BSCS0081_Assignment02.pdf
```

4. Upload it on the **course portal** before **Thursday, 29 Oct 2026, 11:55 PM**.



Illustrative mock-up — the real GitHub screens may look slightly different.

Troubleshooting

“parking.txt not found!”

The .txt is not next to the .vcxproj, its name is spelled differently, or it is really parking.txt.txt.

Error C4996 on strcpy

Add #define _CRT_SECURE_NO_WARNINGS after the header comment, before the #include lines.

Console closes at once

Run with Ctrl + F5, not F5. Do not add system("pause") or getch().

Can’t find the output file

Look in the project folder, next to the .vcxproj. It is created there when the program runs.

“main already defined”

Two .cpp files are in one project. Keep one .cpp per project, one project per question.

Q4 never finishes

A failed read stays failed: call inFile.clear() before skipping junk. Stop the program with Ctrl + C.

For queries

Email

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Office hours

Wednesday 11:00 AM – 1:00 PM
Friday 10:00 AM – 12:30 PM

Office venue

A Building F301 – Cabin 4